LAB ASSIGNMENT-18.3

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Task 1 – Public Transport Fare API

Prompt

"Generate Python code using requests to fetch fare between two stations from a public transport API. Include input validation and error handling."

Code

```
import requests

def get_fare(source, destination):
    if not source or not destination:
        print("Invalid input")
        return

    try:
        url = f"https://api.example.com/fare?from={source}&to={destination}"
        response = requests.get(url, timeout=5)

        if response.status_code == 200:
            data = response.json()
            print("Fare:", data.get("fare", "Not available"))
        else:
            print("API Error:", response.status_code)

    except requests.exceptions.Timeout:
        print("Request timed out")
    except requests.exceptions.RequestException:
        print("Network error")

get_fare("StationA", "StationB")
```

Task 2 – Currency Exchange

Prompt

“Write Python code to convert INR to USD, EUR, GBP using an exchange rate API with validation and error handling.”

Code

```
import requests

def convert_currency(amount):
    if amount <= 0:
        print("Invalid amount")
        return

    try:
        url = "https://api.exchangerate-api.com/v4/latest/INR"
        res = requests.get(url)
        data = res.json()

        usd = amount * data["rates"]["USD"]
        eur = amount * data["rates"]["EUR"]
        gbp = amount * data["rates"]["GBP"]

        print("USD:", usd)
        print("EUR:", eur)
        print("GBP:", gbp)

    except Exception:
        print("Error fetching rates")

convert_currency(1000)
```

Task 3 – GitHub Repository Info

Prompt

“Generate Python function to fetch GitHub repo details (stars, forks, issues) with error handling for 404 and rate limits.”

Code

```
import requests
```

```
def get_repo_info(owner, repo):
    try:
        url = f"https://api.github.com/repos/{owner}/{repo}"
        res = requests.get(url)

        if res.status_code == 200:
            data = res.json()
            print("Name:", data["name"])
            print("Stars:", data["stargazers_count"])
            print("Forks:", data["forks_count"])
            print("Open Issues:", data["open_issues_count"])
        elif res.status_code == 404:
            print("Repository not found")
        elif res.status_code == 403:
            print("Rate limit exceeded")
        else:
            print("Error:", res.status_code)

    except Exception:
        print("Request failed")

get_repo_info("octocat", "Hello-World")
```

Task 4 – News Headlines Aggregator

Prompt

"Create Python script to fetch top 5 news headlines using News API with retry mechanism and error handling."

Code

```
import requests
import time

def get_news(category):
    api_key = "YOUR_API_KEY"
    url = f"https://newsapi.org/v2/top-headlines?category={category}&apiKey={api_key}"

    for i in range(2):  # retry
        try:
            res = requests.get(url)
```

```
        if res.status_code == 200:
            articles = res.json()["articles"][:5]
            for i, a in enumerate(articles, 1):
                print(f"{i}. {a['title']}")
            return
        else:
            print("Error:", res.status_code)

    except Exception:
        print("Retrying...")
        time.sleep(2)

print("Failed to fetch news")

get_news("technology")
```

Task 5 – COVID-19 Statistics API

Prompt

"Write Python code using requests to fetch COVID-19 stats by country with error handling, validation, and retry logic."

Code

```
import requests
import time

def get_covid_data(country):
    url = f"https://disease.sh/v3/covid-19/countries/{country}"

    for i in range(2):
        try:
            res = requests.get(url)

            if res.status_code == 200:
                data = res.json()
                print("Cases:", data["cases"])
                print("Deaths:", data["deaths"])
                print("Recovered:", data["recovered"])
                print("Active:", data["active"])
```

```
            return

        elif res.status_code == 404:
            print("Invalid country")
            return

        elif res.status_code == 429:
            print("Rate limit hit, retrying...")
            time.sleep(2)

    except Exception:
        print("Network error, retrying...")
        time.sleep(2)

print("Failed to fetch data")

get_covid_data("India")
```